**Resume:**


**Ruohan Feng**

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## SUMMARY


Researcher centers on unconscious cognitive and affective processes, with a particular focus on implicit emotion regulation, the neural dynamics of mind-wandering, and intuitive problem solving. I leverage computational modeling to decode these latent states from behavioral and neurophysiological data. Additionally, my work extends to mental health–related mechanisms, including the effects of learned helplessness on psychological flexibility and dysregulated emotional processes.


## EDUCATION

**Georgia Institute of Technology, Atlanta, GA** | GPA: 3.81 *Sep 2023 – Aug 2027 (Expected)*

*Ph.D. Candidate in Cognitive & Brain Science*

*M.S. in Computer Science, specializing in Machine Learning*

Relevant Coursework: Research Method, Computational Data Analysis, Modeling and Simulation, Data and Visual Analytics, Machine Learning, Regression, Statistics, Neuroimage

**Hong Kong Baptist University, Hongkong** *Sep 2014 – Jun 2018*

*B.S. in Applied Psychology*

## WORK EXPERIENCE

**Georgia Institute of Technology, Lab Instructor of Statistics for HCI.** *Jan – May 2025; Jan – May 2026*

- Led weekly labs for Statistics, training about 30 students in **R** across end-to-end analytical workflows, including **data cleaning, visualization, and statistical modeling**.
- Mentored students in applying and interpreting **t-test, ANOVA, ANCOVA, regression, multilevel and hierarchical modeling,** with an emphasis on statistical diagnostics and assumption checking to ensure statistical validity.

**Vanke Co., Ltd. (Vanke Group), Data Operations Intern –Tianjin, China** *Jan 2023 – Aug 2023*

- Translated finance workflow requirements into functional specifications and supported API-based data integration, collaborating with engineering teams on system enhancements.
- Improved financial data standardization and cleansing by **refining mapping logic, adding validation and exception handling**; used SQL to extract structured datasets for **system optimization**, reducing manual reconciliation by ~15–20%.

**Syracuse University, SELF Lab, Research Coordinator, Syracuse, NY** *Jan 2021 – Jun 2022*

- Collected and analyzed a 2,121-participant dataset using multilevel models; built data cleaning and analysis pipelines, contributing statistical insights to a peer-reviewed publication.
  **Publication Link**: https://link.springer.com/article/10.1007/s40688-025-00548-5
- Conducted ERP preprocessing (**resampling, average re-referencing, band-pass filtering, ocular artifact attenuation**) in MNE-Python, extracting amplitudes and testing regulation-condition effects with **repeated-measures ANOVAs.**

## LEADERSHIP EXPERIENCE

**Georgia Institute of Technology**, **Host & Organizer of Brown Bag Academic Seminar** *Sep 2024 – Dec 2025*

- Organized and hosted a weekly departmental research seminar and scholarly presentation forum, coordinating session logistics and moderating each meeting.

## SELECTED PROJECTS

**EEG State Modeling with Graph Features & HMM |** *MNE, NumPy/Pandas, SciPy, NetworkX* *Sep 2025 – Present*

- Built an end-to-end EEG analysis pipeline in Python (MNE, SciPy, NumPy/pandas), extracting multilayer connectivity features (**band-limited correlations, PLV**) and applying **PCA, kNN-based graph construction, Louvain clustering, and Gaussian HMMs** to model latent brain-state transitions and mind-wandering dynamics.
  **GitHub repository**: https://github.com/rfeng77/CSE6242_decode_mind_wandering
- Used **React and JavaScript** to develop interactive visualizations—including spatial-temporal scalp-map animations and an ROI feature-explorer dashboard—to display EEG dynamics, state sequences, and model performance across brain regions.
  **Live Demo**: https://cse6242-project.vercel.app/

**Behavioral Analytics & State-Transition Modeling |** *tidyverse, lme4, afex, tvem, depmixS4, kernlab* *Jan 2024 –July 2025*

- Engineered a trial-level behavioral analytics pipeline in R, integrating robust data cleaning, normalization, and temporal feature construction to quantify reward-modulated changes in cognitive flexibility.
- Leveraged advanced statistical and unsupervised modeling—including **mixed ANOVA, Time-Varying Effect Modeling (TVEM), spectral clustering, and Hidden Markov Models**—to uncover latent behavioral regimes and characterize state-transition dynamics. **PsyArXiv link:** https://osf.io/preprints/psyarxiv/qj2av_v1

## SKILLS

***Programming & Computational Tools:*** R, SQL, Python (PySpark, pandas, NumPy, PyTorch, Matplotlib, TensorFlow), Microsoft Excel, MATLAB, GitHub, Java, LaTeX, SPSS, RevMan, Qualtrics, Tableu, PsychoPy.

***Quantitative & Modeling Expertise:*** Advanced statistical methods (e.g., multilevel modeling, SEM, time-varying effect modeling); Experimental design (e.g., A/B test); Machine learning; Deep learning; Computational modeling (e.g., Bayesian inference, latent growth models Psychometric analysis; Data visualization; Signal processing

**CV:**

**Ruohan Feng**
3154507317| rfeng68@gatech.edu | https://rfeng77.github.io
Atlanta, USA

## EDUCATION

**Georgia Institute of Technology, Atlanta**| GPA: 3.81 *Sep 2024 -Jan 2028 (Expected)*
Ph.D. Candidate in Cognitive & Brain Science, Minor in Machine Learning
M.S. in Computer Science

**Comprehensive (Qualifying) Exams – Completed by September 15, 2025**
*Coursework:* Modeling and Simulation, Computational Data Analysis, Data and Visual Analytics, Machine Learning, Regression, Statistics

**Syracuse University** **Syracuse, USA**
**Clinical Mental Health Counseling, Master of Science** 08/2019-5/2022
*Coursework:* Research Methods, Theories of Counseling, Group Work in Counseling, Foundations of mental health counseling, Crisis Counseling, Psychological Diagnosis, treatment and psychopharmacology, and Internship

**Hong Kong Baptist University** **Zhuhai/Hongkong, China**
**Applied Psychology, Bachelor of Science (Honors)** 09/2014 - 06/2018
*Coursework:* Experimental Psychology, Statistical theory, Computer-Aided Data Analysis, Counseling Psychology, Cognitive Psychology, Psychological Assessment and Testing, and Abnormal Psychology
Dissertation: "The Effect of Anxiety Sensitivity and Its Lower-order Anxiety Sensitivity Factors on Generalized Anxiety Disorder Tendency in China College Students."

## SKILLS

***Programming & Computational Tools:*** R, Python (PySpark, pandas, NumPy, PyTorch, Matplotlib, TensorFlow), MATLAB, SQL, GitHub, Java, LaTeX, PsychoPy, SPSS, RevMan, Qualtrics, Tableu, Microsoft Excel
***Neuroimaging Modalities:*** fMRI, sMRI, EEG
***Neuroimaging Analysis Toolkits:*** FSL, FreeSurfer, ANTs, AFNI, MRIQC, fMRIPrep, Nipype, GLMsingle, GingerALE, SDM, MNE
***Quantitative & Modeling Expertise:*** Machine learning; Deep learning; Computational modeling (e.g., Bayesian inference, latent growth models); Advanced statistical methods (e.g., SEM, multilevel modeling, time-varying effect modeling); Psychometric analysis; Experimental design; Data visualization; Signal processing
***Research & Communication:*** Scientific writing; Research communication; Public speaking; Data interpretation and presentation
***Languages:*** Chinese (native), English (fluent), Japanese (basic)

## PUBLICATION

***Peer-reviewed Publications***

[1]. Ou, Y., Zhang, P., Yu, J., Li, M., Su, S., Zhang, M., **Feng, R.,** Sun, F., & Wu, J. (2025). *The application of the BERTopic model in natural language processing: In-depth text topic modeling*. In Proceedings of the 2025 5th International Conference on Consumer Electronics and Computer Engineering (ICCECE) (pp. 793–796). IEEE. https://doi.org/10.1109/ICCECE65250.2025.10984639

[2]. **Feng, R.**, Mishra, V., Hao, X., & Verhaeghen, P. (2024). The Association Between Mindfulness, Psychological Flexibility, and Rumination in Predicting Mental Health and Well-Being Among University Students Using Machine Learning and Structural Equation Modeling. *Machine Learning with Applications*, 100614. https://doi.org/10.1016/j.mlwa.2024.100614

[3]. Razza, R., Liu, Q., **Feng, R.** Hao., X., Kirkman, K., & Merrin, G. J. (in press). Cultivating adolescents' self-compassion through mindfulness: The role of self-regulation at both the individual- and classroom-level. *Contemporary School Psychology*.

[4]. Wang, Q., Wu, Y., **Feng, R.,** Hao, X., Felver, J. C., Zhang, Y., & Razza, R. (2024). Adaptation and Validation of the Mindful Student Questionnaire in Chinese. *Mindfulness*, *15*(2), 359-371. https://doi.org/10.1007/s12671-023-02299-x

[5]. Lee, P., Chen, S., Yu, Y., Chen, Y., **Feng, R**., & Fan, X. (2019). The Effect of Mindfulness and Wisdom as Intervention to Deal with College Students' Internet Addiction. *International Journal of Psychology & Behavior Analysis. 5*, DOI: https://doi.org/10.15344/2455-3867/2019/161

***Manuscripts Under Review***

[6]. **Feng, R.,** Yin, S., Jahanzad, S., Hao, X., Apt, L., & Verhaeghen, P An Experimental Investigation of Implicit Emotional Counter-Regulation and Its Neural Correlates Using a Priming Identity Task (*in Revision of Cognitive, Affective & Behavioral Neuroscience)*

[7]. Hao, X., **Feng, R.,** Guimarães, A. L., Verhaeghen, P., & Turnbull, A. G. Mind–body interventions and brain changes in aging adults with mild cognitive impairment: A systematic review of structural and functional neuroimaging findings. (*in Revision of Neuropsychology Review*)

[8]. **Feng, R**., Hao, X., Coronado, B., & Verhaeghen, P. When Control Slips Away: Temporal Dynamics of Learned Helplessness and Cognitive Flexibility under Reward Uncertainty *(in Revision of Motivation and Emotions)*
**PsyArXiv link:** https://osf.io/preprints/psyarxiv/qj2av_v1

**[9]. Feng, R.**, Hua, Z., Chen, P., Lin, T., & Veldhuis, C. A novel application of network analyses to examine how well LGBTQ research in LGBTQ journals reflects the diversity of the community (in Revision of *Annals of Behavioral Medicine*)

## CONFERENCE

[1]. **Feng, R.,** Yin, S., Jahanzad, S., Hao, X., Apt, L., & Verhaeghen, P An Experimental Investigation of Implicit Emotional Counter-Regulation and Its Neural Correlates Using a Priming Identity Task, SANS (Social & Affective Neuroscience Society), San Diego, USA

[2]. Hao, X., **Feng, R.,** Guimarães, A. L., Verhaeghen, P., & Turnbull, A. G. (2025) Mind–body interventions and brain changes in aging adults with mild cognitive impairment: A systematic review of structural and functional neuroimaging findings. 2025 International Society for Contemplative Research, North Carolina, USA

**[3]. Feng, R.**, Hua, Z., Chen, P., Lin, T., & Veldhuis, C. A novel application of network analyses to examine how well LGBTQ research in LGBTQ journals reflects the diversity of the community, 2, Minneapolis, USA, 2022.

[4]. **Feng, R**. The Effect of Anxiety Sensitivity and Its Lower-order Anxiety Sensitivity Factors on Generalized Anxiety Disorder Tendency in China College Students. Singapore Conference on Applied Psychology (SCAP), Singapore. 2019

[5]. **Feng, R.** The Effect of Delay Discounts on College Students with Online Shopping Addiction: Learned Helplessness as a Mediator Variable. Asian Conference on Psychology & the Behavioral Sciences, Tokyo, Japan. 2018

## ONGOING PROJECTS

[1]. Human-AI Interaction in Medical and Psychological Consultation: Modeling Behavior and Decision-Making

[2]. Toward Generalizable Mind-Wandering Decoding: Integrating Shapelet and Coherence Features via Variational Autoencoder (Ongoing project since 2025)

[3]. Mindfulness and Creative Problem-Solving with Sudden Insight: evidence from an electrophysiological study (Ongoing project since 2024)

[4]. An Ecological Momentary Assessment (EMA) Study of Mind-Wandering, Affect, and State Mindfulness (Ongoing project since 2023)

## WORK EXPERIENCE

**Georgia Institute of Technology, Lab Instructor (Psychological Statistics for HCI)** *Jan – May 2025; Jan – May 2026*

- Led weekly labs for Statistics, training about 30 students in **R** across end-to-end analytical workflows, including **data cleaning, visualization, and statistical modeling**.
- Mentored students in applying and interpreting **t-test, ANOVA, ANCOVA, regression, multilevel and hierarchical modeling,** with an emphasis on statistical diagnostics and assumption checking to ensure statistical validity.

**China Vanke Co., Ltd. (Vanke Group), Data Operations Intern –Tianjin, China** *Jan 2023 – Aug 2023*

- Translated finance workflow requirements into functional specifications and supported API-based integration for a third-party financial data platform.
- Improved financial data standardization, mapping, and cleansing workflows by refining mapping logic, adding validation and exception handling, and coordinating with engineering and delivery teams to enhance data consistency and reduce manual reconciliation efforts by an estimated ~15–20%.

**Institute of Economics of Education, Peking University** Beijing, China
Research Assistant (Part-Time)| Supervisor: Dr. Hong Zhu 10/2022 – 05/2023

- Supported a longitudinal evaluation of lifelong education programs using behavioral and EEG data. Assisted with experimental paradigm design and preprocessing of neurophysiological signals.

**SELF Lab-Syracuse University, Research Coordinator, Syracuse, NY**
*Jan 2021 – Jun 2022*

- Analyzed a 2,121-participant mindfulness dataset using hierarchical multilevel models; built pipelines for large-scale cleaning, centering, model comparison, and variance-partitioning, contributing statistical insights to a peer-reviewed publication. **Publication Link**: https://link.springer.com/article/10.1007/s40688-025-00548-5
- Conducted ERP preprocessing (**resampling, average re-referencing, band-pass filtering, ocular artifact attenuation**) and analysis in MNE-Python, extracting N170/EPN/LPP amplitudes and testing regulation-condition effects with **repeated-measures ANOVAs.**

**Mind-Body Lab, Syracuse University** (volunteer, unpaid position) Syracuse, USA
Research Assistant | PI: Dr. Joshua Felver 08/2021 – 06/2022

- Translated, collected, and analyzed data from the Mindful Student Questionnaire with 2,910 international adolescents to examine mindfulness factors (awareness, attention, acceptance) influencing school performance; co-authored a peer-reviewed publication in Mindfulness [4].
- Conducted TSST tests to aid graduate students in conducting experiments, including blood pressure measurements and response time data collection (2–3 times per week).

## TEACHING EXPERIENCE

PSY 2012 – Research Methods, Instructor, 2026 Summer
PSY 2020 – Psychological Statistic, Lab instructor, 2026 Spring
PSY 2020 – Psychological Statistic, Lab instructor, 2025 Spring
PSY 2020 – Psychological Statistic, Teaching assistant, 2024 Fall
PSY 1101 – General Psychology, Teaching assistant, 2024 Summer
PSY 2015 – Research Methods, Teaching assistant, 2024 Spring
PSY 3012 – Introduction to Cognitive Psychology, Teaching Assistant, 2023 Fall

## ACADEMIC SERVICE

Coordinator for the Brown Bag Academic Presentation Series, School of Psychology, Fall 2024 – Spring 2025